

MOHAMMED HALBOUP

Saarbrücken, Germany

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PROFILE

Full-Stack Engineer (Werkstudent) at PCC and Computer Science student at Saarland University, focused on building reliable software systems and shipping production-ready features.

EDUCATION

B.Sc. Computer Science Universität des Saarlandes, Saarbrücken, Germany 2024 – Present
Cambridge A-Levels (CAIE) Computer Science (A*), Mathematics (A*), Physics (A*) 2022 – 2023

EXPERIENCE

- PersonalClientCare (PCC)** September. 2025 – Present
Full-Stack Engineer (Werkstudent) Germany / Remote
- Developed user-facing UI features and backend automation for application.
 - Designed and implemented backend APIs and data workflows supporting intelligent automation.
 - Delivered scalable, production-ready improvements in an agile engineering environment.
- Universität des Saarlandes** August 2026 – Present
Software Engineering Lab Teaching Assistant, Prof. Dr. Sven Apel Saarbrücken, Germany / On-site
- Architected and developed **Saar la carte**, a comprehensive restaurant simulation platform with end-to-end farming workflows, including environment management, customer interactions, order processing, service delivery, and incident handling.
 - Mentored students through the complete software engineering lifecycle, providing guidance on system design, implementation, testing strategies, and debugging techniques across multiple project phases.
 - Designed and implemented comprehensive validation, component, and scenario tests that serve as course assessment criteria, ensuring code quality and proper software engineering practices.

PROJECTS

- MetalPulse** — GitHub — Preview
- Real-time precious metals dashboard with live spot prices, ETF tracking, price alerts, market news, and an AI-powered market analyst chat.
- InkedIn** — GitHub — Preview
- Built a full-stack finance website with Flask, HTML, CSS, and JavaScript.
- SwissInsight** — GitHub — Preview
- Natural language query system for Swiss startup data using SPARQL and semantic search.
- Klebl AI (Hackathon Winner)** — GitHub
- Used YOLOv11 and Gemini API to analyze construction blueprints, reducing analysis time by 90%.
- DreamTransport** — GitHub
- Public transport optimization system for Neumarkt i.d. OPf. to improve bus routes and stops using passenger data, punctuality analysis, and mathematical optimization algorithms.
- UdS_OP (Universal Desktop Assistant)** — GitHub
- AI assistant capable of code execution, file operations, browsing, and searches.
- Voice Oasis** — GitHub
- Voice enhancement platform for AI agents using SpeechBrain, Azure Cognitive, and Flask.

TECH SKILLS

Python, C, SQL, Go, Dart, HTML, CSS, JavaScript, OCaml, Scala, Flask, FastAPI, Flutter, RISC-V, Chisel, Git, GitHub, VS Code, Google Workspace, Microsoft Office, Adobe Creative Suite

LANGUAGES

Arabic: Native, **English:** C1 Proficiency, **German:** B1 Proficiency